EPILOGUE

Everything Is Quantum

What would Einstein have thought of Many-Worlds quantum theory? Likely he would have been repulsed, at least at first exposure. But he would have to admit that there are aspects of the idea that fit very well with his picture of how nature should operate.

Einstein died in Princeton in 1955, just as Everett was wrangling his idea into shape. He was firmly committed to the principle of locality, and was enormously bothered by the spooky action at a distance implied by quantum entanglement. In that sense, he might very well have been horrified by Many-Worlds and the holographic principle, ideas that treat space itself as emergent rather than fundamental. The suggestion that reality is described as a vector in an enormous Hilbert space, rather than as matter and energy in good old four-dimensional spacetime, is not one he would have found congenial. But there's a good chance that he would have been pleased that Everett returns our best description of the universe to one featuring definite, deterministic evolution—and reaffirms the principle that reality is ultimately knowable.

Late in life, Einstein related a story from his childhood.

> A wonder of this kind I experienced as a child of four or five years when my father showed me a compass. That this needle behaved in such a determined way did not at all fit in the kind of occurrences that could find a place in the unconscious world of concepts (efficacy